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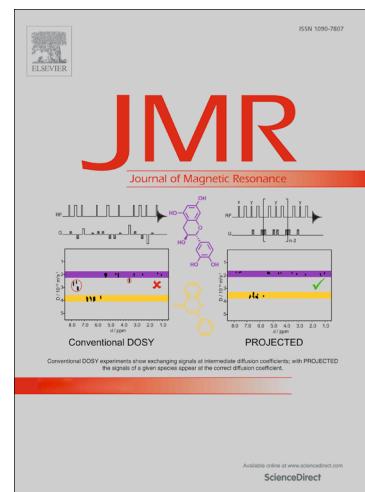
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A Low-cost home-built NMR using Halbach magnet

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Abstract: The objective of this work is to develop a low-cost compact desktop NMR system based on Halbach magnets with the advantages of small size and ability to generate relatively high field strength. Considering the cost of manufacturing and assembling the magnetic blocks, the system utilized a 3-layer Halbach magnet and a wedge-shaped mechanical structure, which was designed for magnet rapid assembling. The comparison between simulation and calculation results of the initial magnetic field strength distribution showed that design theory and practice were in accordance. The initial homogeneity was 576 ppm in a square with a length of 5 mm. After passive shimming with two magnetic blocks and steel pieces, the uniformity reached 120 ppm in the same area. We developed and tested a compact single board spectrometer with digital modulation and demodulation in order to enhance the system mobility and improve the SNR. A self-made probe was used to carry out experiments with the spectrometer, and the spectral width at half-height reached 20 ppm in a cylinder with a diameter of 1.5 mm and a length of 1 mm. Compact structure and low cost of the system will facilitate and extend the application of desktop NMR system.

Keywords: Desktop NMR; Halbach magnet; Digital spectrometer.

1. Introduction

NMR (Nuclear Magnetic Resonance) is widely used in industry and food testing, prospecting, quality control and macromolecular structure analysis [1-5]. NMR systems are usually based on superconducting magnets with field strength up to 11 Tesla and a weight of several tons. Among the advantages of superconducting NMR are high field strength, good resolution and sensitivity. However, the drawbacks of the superconducting NMR are also obvious, such as bulky size and a high cost of maintenance. The widespread application of NMR in biology and industrial fields is limited because they are hard to integrate to clinical *in vivo* cell, tissue research and quality control in factories [2, 6]. Desktop NMR systems have already been developed for covering the shortage of superconducting NMR. Permanent magnets are usually used in desktop NMR for its small size and low cost, while the homogeneity of the permanent magnet cannot be compared with that of superconducting magnets. Desktop NMR has more compact size and lower cost than superconducting NMR but has worse resolution and sensitivity. The design of the permanent magnet of high homogeneity for desktop NMR has been developed in recent years to resolve these limitations [2, 7-29]. Moreover, the application of desktop NMR has made a great

progress and has been successfully utilized in chemistry and medical study [3, 4, 6, 30].

The desktop NMR system mainly relies on a homogeneous and high field magnet. Unilateral magnets and Halbach magnets are mostly used in desktop NMR. A unilateral magnet can detect samples outside its volume but the homogeneous area is limited and it can also barely achieve high field strength. Due to its capacity for generating higher magnetic field strength, Halbach magnets are always the research focus of desktop NMR [2, 31]. The Halbach magnet has the largest field strength-mass ratio among all types of magnets and has no external stray fields without the use of yoke [25]. However, a Halbach magnet consists of several or tens of magnetic blocks with different magnetizations angles and poses requirements on manufacturing precision. In particular, the design of high homogeneous Halbach magnets lacks enough theoretical support and it is harder to shimming compared with other types of magnets.

In recent years, the developments of desktop NMR and Halbach magnet have achieved great progress. Moresi and Magin proposed a 0.6 T Halbach magnet for desktop NMR with homogeneity better than 20 ppm. The spectrometer for the preliminary study was constructed by isolated integrated circuits and LabVIEW acquisition board [8]. In 2010, Danieli and Perlo released a small NMR system using a 3-layer Halbach permanent magnet with a 0.7 T field strength and field deviations of 0.15 ppm [16]. Soltner and Blümli presented formulas and numerical values to

calculate the flux density of a single Halbach ring, and their research was based on a dipole approach which is similar to the calculation for the optimum construction of a Helmholtz coil [18].

Compared with the shimming of a C-type or an H-type permanent magnet, the shimming of a Halbach magnet is harder due to its relatively smaller inner space. Some methods were proposed: positioning additional smaller permanent magnetic blocks, using another magnet ring placed at the interior or exterior of the magnet and moving magnet radially inward or outward as reported in earlier papers [15, 28, 32-35]. Most of the shimming schemes require the precise mechanical position in radial or axial directions, particularly for magnetic blocks with different magnetization angles and also require shimming experience. Moreover, shimming of a Halbach magnet means a lot of costs and effort.

As one of the most critical elements in an NMR system, the spectrometer generally has the ability to generate radio frequency (RF) pulse, receive signals from RF coils and process the NMR data. For desktop NMR, commercial spectrometers are generally equipped with closed software and hardware and are difficult to integrate [36]. Most desktop NMR systems described in the literature mainly use commercial spectrometers or isolated components to set up the system [37-44]. To improve the signal-noise-ratio (SNR), different spectrometer components were proposed in several pieces of research but no complete miniature NMR spectrometer has been issued [2, 45].

In this paper, we aim to develop a low-cost and compact home-built NMR system. A 3-layer Halbach magnet was designed and manufactured based on our previous work [26]. The passive shimming was implemented by placing magnetic and steel pieces on the inner face of a bracket assembled in the Halbach magnet to reduce the cost and effort of shimming. In addition, a compact spectrometer with digital modulation and demodulation was also designed to be integrated with the system. Hall sensor and self-made probe were used to measure the uniformity of the magnet, and the SNR of the receiver was also tested and compared with the existing spectrometer.

2. Methods

2.1 Magnet and Shimming

The 3-layer Halbach magnet has 16 segments at each layer, as shown in Fig.1. The simulation result of the homogeneity was 45 ppm in a 5 mm diameter sphere [26]. Usually, the multilayers Halbach magnet is hard to install due to the force between magnets and because the magnetic material is very fragile. A novel method for rapid installation and positioning of magnetic blocks was proposed. Trapezoidal shapes and small trapezoidal tails of magnetic blocks were used for installation and positioning. It can hardly attain a good homogeneity below 10^{-5} in the 10% of inner diameter even with precise mechanical position and identical magnetic blocks, which mainly have variations of magnetizing angles and magnetization since the homogeneity is affected

by many factors including imperfect magnetic material blocks, manufacturing and assembling errors [31]. The magnetizing angle error of which the Halbach magnet we built was below 3 degrees and the magnetization errors were on the order of 10^{-3} .

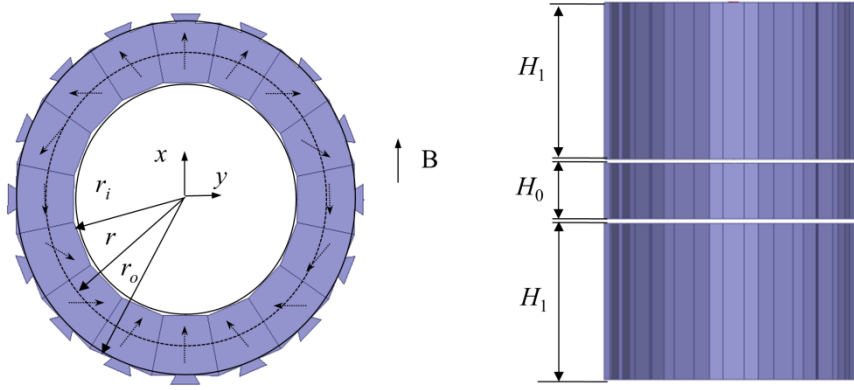


Fig.1 A 3-layer Halbach cylinder with 16 segments at each layer. Each magnet block has a trapezoidal shape and a small trapezoidal tail, which are used for installation and positioning.

A low-cost shimming manner was applied to our magnet by splicing small magnets or steel pieces on the surface of the grooves of the inner wall of an aluminum barrel. The x component of the magnetic field in a Halbach magnet could be expressed as the sum of spherical harmonics series as in equation (1) [46].

$$B(r, \theta, \phi) = \sum_{n=0}^{\infty} \sum_{m=0}^n r^n P_n^m(\cos \theta) (A_n^m \cos m\phi + B_n^m \sin m\phi) \quad (1)$$

The shimming process could be described as in equation set (2), where n is the number of points in the interesting magnetic field and m is the number of positions in which steel pieces could be placed. B_j is the magnetic field strength of j point and B_j is the corrected magnet field strength of a point j . ΔB_{ji} is a feature matrix used to

describe the influence of the magnet field at different points j due to the steel piece at position i . ΔB_{peak} is the maximum magnetic field difference in the interesting zone. B_{max} is the maximum field strength while B_{min} is the minimum field strength of B_j vector. M_i is the number of steel pieces at shimming position i . The objective of the optimization was to find the best magnet pieces matrix M_i to get the minimal value of ΔB_{peak} . ΔB_{ji} matrix can be obtained by simulation and measurement. The algorithm could be executed iteratively for better results.

$$\begin{cases} B_{j'} = \Delta B_{ji} M_i + B_j, i = 1, 2, \dots, m, j = 1, 2, \dots, n \\ \Delta B_{peak} = B_{max} - B_{min} \\ Min(\Delta B_{peak}) \end{cases} \quad (2)$$

2.3 Digital spectrometer

A compact spectrometer is essential for desktop NMR due to the space limitation. Commercial spectrometers like Magritek, Kea, and MRI solutions are widely used in desktop NMR systems. In an NMR system, the spectrometer generates a certain frequency RF pulse, which is transmitted to the coil to make nuclei to resonate and receives the NMR signal for the digitalization. The NMR signal needs analog down-conversion, filtering, amplification and digitalization before it can be sent to the computer. As displayed in Fig.2, the functional block of the spectrometer mainly contains a transmitter, a receiver, and a control module. An FPGA (Xilinx, San Jose, CA) was used to mainly realize the sequence control, NMR data receiving, digital down-conversion and digital filtering. The sequence data was transmitted to the spectrometer and stored in the FPGA RAM. When the NMR scan sequence start

command was emitted, RF wave data were transmitted to AD9957 (Analog Device Inc.) with a frequency digital upconversion. The NMR signal firstly passed through a self-made low noise preamplifier (LNA) of which the gain was 30 dB and the noise figure was 0.5 dB. Then the amplified signal was sent to a variable gain amplifier (VGA) ADL5202 (Analog Device Inc.) with a gain of -12~25 dB. After band-pass filtering, the signal was directly digitalized by a 125 MSPS ADC (AD9653, Analog Device Inc.), then the digital signal flowed into an FPGA digital frequency down-conversion module which was based on DDS (Direct Digital Synthesis), and was filtered by CIC filters in the FPGA. The spectrometer exchanged data with the computer user interface over a 1 Gbps Ethernet.

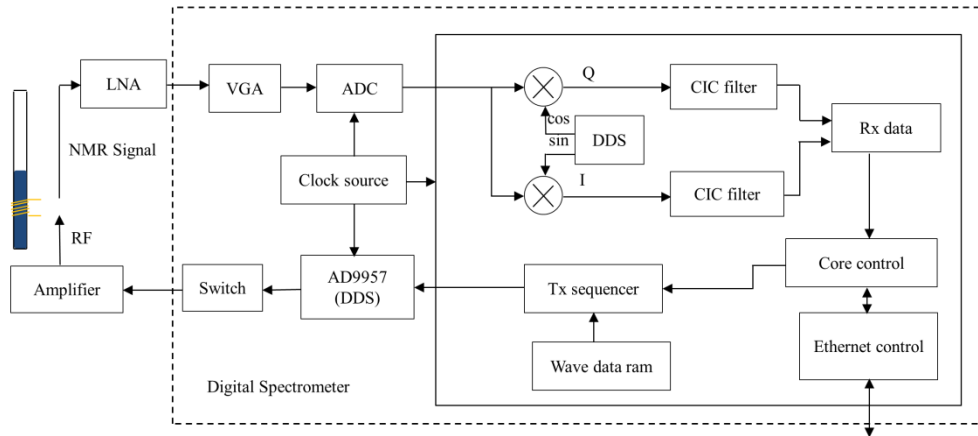


Fig.2 Spectrometer electrical functional block

3. Results

The 3-layer Halbach magnet consisted of 48 NdFeB-42M blocks as shown in Fig.3. The inner radius r_i was 39 mm and the outer radius r_o was 59 mm. The height of the upper and lower layer H_1 was 70 mm while the central layer height H_0 was

25.179 mm. The gap between layers was 2.59 mm. The magnetic blocks were installed in a copper barrel with 16 trapezoidal slots, as indicated in Fig.4 (a). The assembling of the 3-layer Halbach magnet could take place within 30 minutes. The B_r of the magnetic blocks was 1.32-1.324 T and the corresponding coercive force was about 998.22 kA/m. The remanence errors of magnetic blocks ranged from 0.1% to 3% while the magnetism angular errors ranged from 0.56 to 3 degrees. The initial profile of the magnet field was measured with the high precision digital gauss meter CH-3600 (CH-HALL Electronic devices Inc. Beijing, China). The magnetic field strength profile on the z-axis was measured automatically with a self-made 3 axes movement robot. The simulation in Fig.4 (b) showed the actual magnetic profile and that the center with the best homogeneity was at $z/r = -0.18$. Additionally, the measured magnetic field was 2% smaller than that of the simulation. The theoretically calculated field strength was 0.533 T that was 6% larger than the actual value. The difference between simulation and reality lay within the acceptable limit.

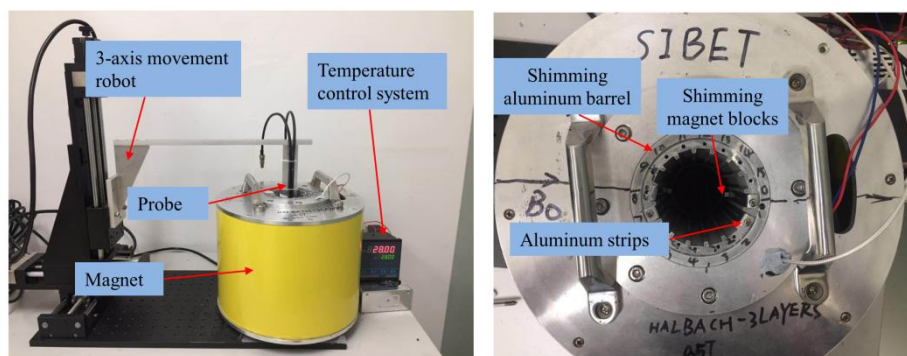


Fig. 3 Magnet Shimming platform with a temperature control system.

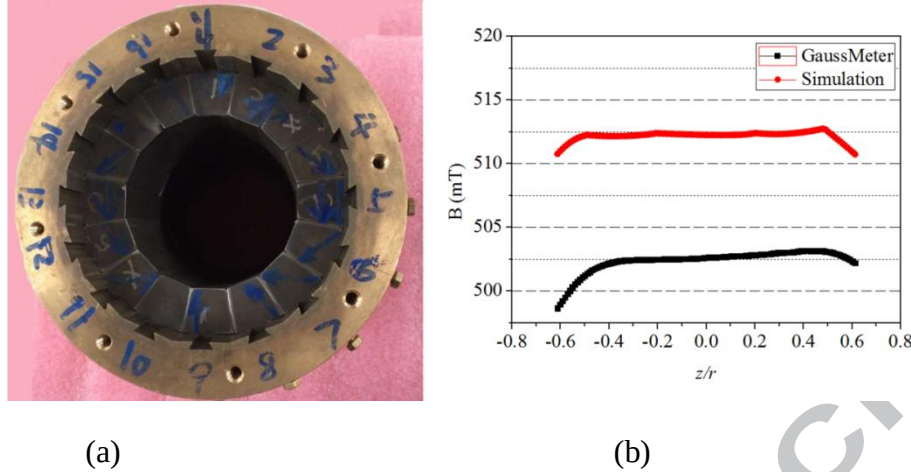


Fig.4. (a) 3-layer Halbach magnet after assembling with a copper shell which has 16 trapezoidal slots. (b) Initial magnetic profile of magnet field on the z-axis. The red line curve represents the simulation of the magnetic field profile. The black line curve denotes the profile measured by Gauss meter.

The shimming was performed with an aluminum barrel which had 16 slots on the inner surface as shown in Fig.5. The shimming strip could be installed in one of the slots of the aluminum barrel. The $5 \times 10 \times 0.2$ mm tungsten steel pieces could be bonded to 5 different places of an aluminum strip and were symmetrically distributed at constant intervals of $d = 10$ mm around the center. As in the equation set (2), shimming steel pieces could be placed at 80 different positions on aluminum strips. The 40 measured points were evenly distributed on the surface of a 5 mm diameter sphere. Matrix ΔB_{ji} was measured with Gauss meter while the shimming aluminum strip was installed from the first to the 16th slot.

The initial magnetic field strength profile ($z = 0$ plane) was measured with a Gauss meter as shown in Fig.6 (a) and the uniformity in the 5 mm long square was

576 ppm. In the x -direction (B_0 direction), the magnetic field gets stronger as it moved farther away from the center but weaker in the y -direction. The magnetic field profile was caused by the limited length, magnetic blocks variations and assembling errors. The magnetic field profile at the center was similar to our theoretical calculated results [23]. Nevertheless, based on the measured ΔB_{ji} , the strip with tungsten steel piece at all 16 slots led to a worse homogeneity on the $z = 0$ plane because the shimming strips with steel piece at all slots enhanced the magnetic field strength more in the x -direction than in the y -direction. The shimming effect of the Halbach magnet with steel pieces was not enough. Hence, a pair of $5 \times 5 \times 10$ mm N35 magnetic blocks was utilized to weaken the field strength in the x -direction. The new magnetic field strength profile was gradient-distributed in the y -direction, and the homogeneity of a square with a length of 5 mm was 350 ppm, as indicated in Fig.6 (b). The shimming algorithm equation set (2) could be executed iteratively several times in order to find M_i . After shimming with steel pieces, the homogeneity of the square could reach 120 ppm, as shown in Fig.6 (c). Finally, a self-made probe was used to evaluate the homogeneity more accurately. The probe was made of a capillary with an inner diameter of 1.5 mm and a length of 1 mm, the coil was made of 8 turns copper wire with a diameter of 0.5 mm as shown in Fig.7 (b). The half-height of the spectral line width could attain 20 ppm. The magnet was assembled in an aluminum shell with a temperature control system, which had the control precision of 0.01 K. The magnet

field drift was 50 ppm in 2 hours(25 ppm/h) according to the measurements performed by our self-made probe.

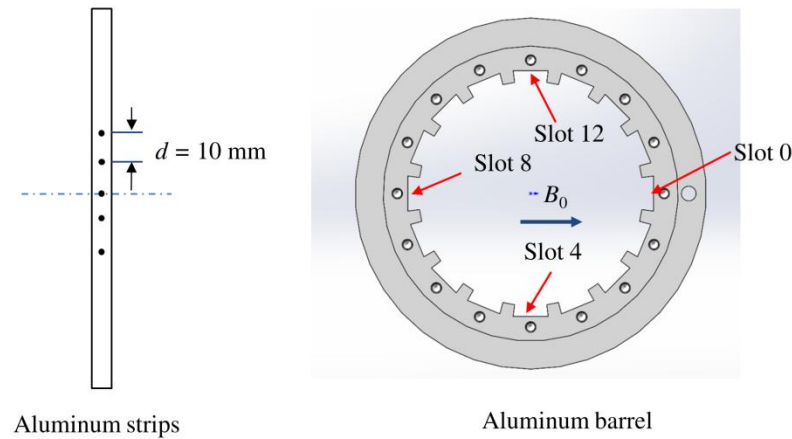


Fig. 5. The shimming aluminum strip was bonded with the $5 \times 10 \times 0.2$ mm tungsten steel pieces. The aluminum barrel with 16 slots used to place shimming strips was precisely installed on the magnet shell.

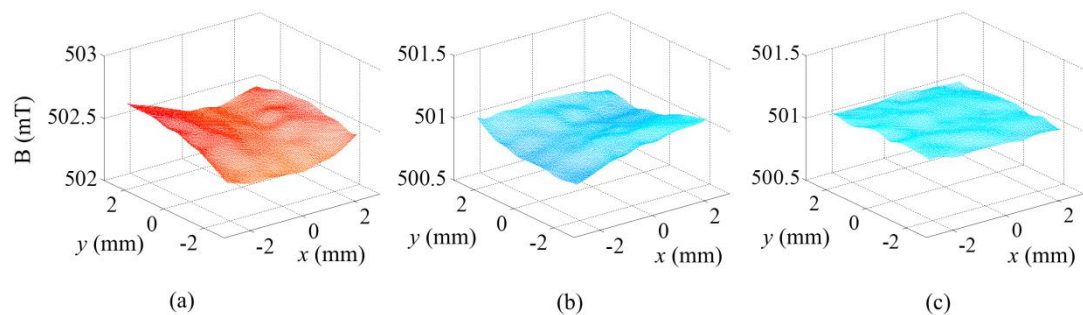


Fig. 6. (a) Initial magnetic field profile at $z = 0$ of a 5 mm long square. The homogeneity was 576 ppm. (b) Magnetic field profile after correction with opposite direction magnetic blocks. The homogeneity was 350 ppm. (c) Magnetic field profile after shimming with steel pieces. The homogeneity was 120 ppm. The color map scale is the same in (a), (b) and (c).

The receiver of the single board spectrometer, as shown in Fig.7 (a), was tested with a DG4102 function waveform generator (RIGOL, China). A 21 MHz sinus wave signal of -1 dBFS amplitude was fed into the receiver, which has the SNR of 65.84 dB, an SNRFS of 67.09 dB and SINAD of 60.493 dBc. The ADC specification indicates that the SNR should be 76 dB. The measured noise floor of FFT spectral was larger than the ADC specification since the jitters of the clock and power supply introduced more noise. An FID sequence pulse was operated with our digital spectrometer. The average SNR was 17 after 5 times. The time domain waveform after demodulation and frequency spectrum was demonstrated in Fig.8. The length of the coil was 1 mm and the volume of the water sample was about 1.8 μ L. The mass sensitivity was 0.107 $S/N \cdot \mu\text{mol}^{-1} \cdot \text{s}^{-1/2}$ and was calculated with the equation (3) [47].

$$S_m = \frac{S/N}{\text{mol} \cdot t_{acq}^{1/2}} \quad (3)$$

An FID sequence with the same parameters was also operated on a commercial spectrometer (DRX2, Oxford Instruments, Oxfordshire, UK). From the frequency spectrum, the SNR was evaluated to 13. The SNR of our home-made digital spectrometer can be improved by 4 since an analog mixer to implement down-convert of frequency in DRX2 may introduce more noise.

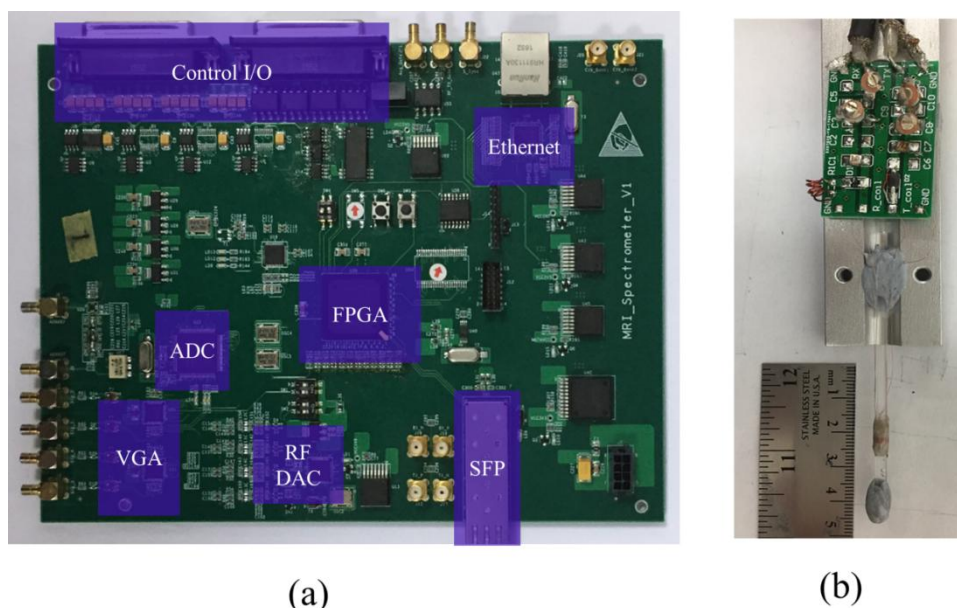


Fig.7. (a) Single board digital spectrometer. SFP, small form-factor pluggable sockets was reserved to be utilized near magnet circumstance. (b) Self-made probe. For better isolation and SNR, the transmit and receive coils were used separately in the probe. And passive mismatch circuit was utilized in the receiver.

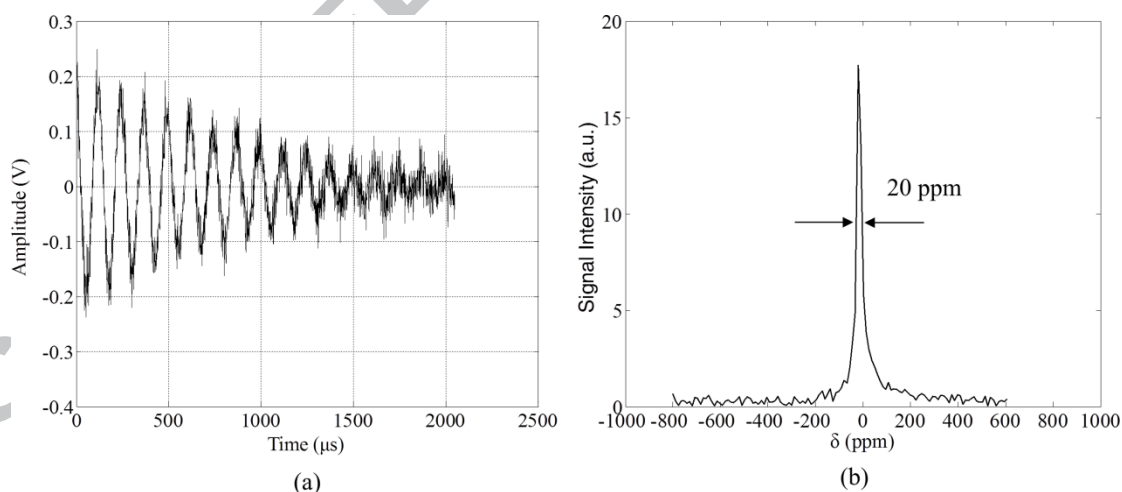


Fig.8. FID sequence was applied in our home-made NMR system with a water sample. (a) Time domain waveform after the frequency digital demodulation. (b) The frequency spectrum of the water sample. The pulse width is 24μ s and the homogeneity reaches 20 ppm.

4. Conclusions

This work proposes a compact desktop NMR system based on a 3-layer Halbach magnet. The multilayer structure of the Halbach magnet can compensate for the inhomogeneity on the z-axis. However, the uniformity of the central plane is also a big problem. After passive shimming with magnetic blocks and steel pieces, the homogeneity of a square with a length of 5 mm was improved from 576 ppm to 120 ppm. Finally, the homogeneity was measured with a self-made probe that could reach 20 ppm without active shimming. While the previous shimming work of researchers' mainly adopted additional magnetic blocks to generate a new magnetic field profile, in order to compensate for the uniformity of the original magnetic field, Danieli's work on the Halbach magnet based on moving the magnetic blocks at radial direction is amazing, and could also be seen as a variation of previous method [16]. Nevertheless, it needs precise mechanical assembling and movement control. We try to reduce the cost and effort by placing magnetic blocks and steel pieces for shimming. Even with good temperature control, the field temperature drift was 25 ppm per hour. The temperature change was smaller than 0.02 K in the NMR system, thus was acceptable for the temperature control system. Therefore frequency lock coils are needed or temperature drift cancellation magnetic blocks should be used [16]. A digital spectrometer was designed for compact structure and better SNR. The successful realization of single board digital spectrometer indicated that this could improve SNR and be applied in a wider field strength range. In the future, the studies

of easy and low-cost shimming methods will be of great importance to achieve better homogeneity. The total cost of the system was estimated at about \$5,500 while the weight was below 35 kg (the magnet is about 25 kg) without considering the RF amplifier. Compact structure and low cost of the system are strong points of our desktop NMR system.

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References

- [1] B. Blümich, F. Casanova, S. Appelt, NMR at low magnetic fields, *Chemical Physics Letters*, 477 (2009) 231-240.
- [2] S.S. Zalesskiy, E. Danieli, B. Blümich, V.P. Ananikov, Miniaturization of NMR systems: desktop spectrometers, microcoil spectroscopy, and "NMR on a chip" for chemistry, biochemistry, and industry, *Chemical reviews*, 114 (2014) 5641-5694.
- [3] B. Blümich, K. Singh, *Desktop NMR and Its Applications From Materials Science To Organic Chemistry*, Angewandte Chemie International Edition, (2017).
- [4] J. Perlo, E.V. Silletta, E. Danieli, G. Cattaneo, R.H. Acosta, B. Blümich, F. Casanova, Desktop MRI as a promising tool for mapping intra-aneurismal flow, *Magnetic resonance imaging*, 33 (2015) 328-335.
- [5] M. Li, C.C. Vassiliou, L.A. Colucci, M.J. Cima, (1)H nuclear magnetic resonance (NMR) as a tool to measure dehydration in mice, *NMR in biomedicine*, 28 (2015) 1031-1039.
- [6] E. Danieli, J. Perlo, A.L. Duchateau, G.K. Verzijl, V.M. Litvinov, B. Blümich, F. Casanova, On-line monitoring of chemical reactions by using bench-top nuclear magnetic resonance spectroscopy, *Chemphyschem : a European journal of chemical physics and physical chemistry*, 15 (2014) 3060-3066.
- [7] S.M. Wright, D.G. Brown, J.R. Porter, D.C. Spence, E. Esparza, D.C. Cole, F.R. Huson, A desktop magnetic resonance imaging system, *Magma*, 13 (2002) 177-185.

- [8] G. Moresi, R. Magin, Miniature permanent magnet for table-top NMR, *Concepts in Magnetic Resonance*, 19B (2003) 35-43.
- [9] H. Raich, P. Blümmler, Design and construction of a dipolar Halbach array with a homogeneous field from identical bar magnets: NMR Mandhalas, *Concepts in Magnetic Resonance Part B: Magnetic Resonance Engineering*, 23B (2004) 16-25.
- [10] W.H. Chang, J.H. Chen, L.P. Hwang, Single-sided mobile NMR with a Halbach magnet, *Magnetic resonance imaging*, 24 (2006) 1095-1102.
- [11] J. Perlo, F. Casanova, B. Blümich, Single-sided sensor for high-resolution NMR spectroscopy, *Journal of magnetic resonance*, 180 (2006) 274-279.
- [12] V. Demas, J.L. Herberg, V. Malba, A. Bernhardt, L. Evans, C. Harvey, S.C. Chinn, R.S. Maxwell, J. Reimer, Portable, low-cost NMR with laser-lathe lithography produced microcoils, *Journal of magnetic resonance*, 189 (2007) 121-129.
- [13] J.C. Perlo, F.; Blümich, B., Ex Situ NMR in Highly homogeneous Fields- ¹H spectroscopy (2007).
- [14] B. Manz, M. Benecke, F. Volke, A simple, small and low cost permanent magnet design to produce homogeneous magnetic fields, *Journal of magnetic resonance*, 192 (2008) 131-138.
- [15] E. Danieli, J. Mauler, J. Perlo, B. Blümich, F. Casanova, Mobile sensor for high resolution NMR spectroscopy and imaging, *Journal of magnetic resonance*, 198 (2009) 80-87.
- [16] E. Danieli, J. Perlo, B. Blümich, F. Casanova, Small magnets for portable NMR spectrometers, *Angewandte Chemie*, 49 (2010) 4133-4135.
- [17] C. Hugon, F. D'Amico, G. Aubert, D. Sakellariou, Design of arbitrarily homogeneous permanent magnet systems for NMR and MRI: theory and experimental developments of a simple portable magnet, *Journal of magnetic resonance*, 205 (2010) 75-85.
- [18] H. Soltner, P. Blümmler, Dipolar Halbach magnet stacks made from identically shaped permanent magnets for magnetic resonance, *Concepts in Magnetic Resonance Part A*, 36A (2010) 211-222.
- [19] W. Liu, F. Casanova, B. Blümich, D. Zu, An Efficacious Target-Field Approach to Design Shim Coils for Halbach Magnet of Mobile NMR Sensors, *Applied Magnetic Resonance*, 42 (2011) 101-112.
- [20] L. Liu, Designing magnets with prescribed magnetic fields, *New Journal of Physics*, 13 (2011) 033013.
- [21] C.W. Windt, H. Soltner, D. van Dusschoten, P. Blümmler, A portable Halbach magnet that can be opened and closed without force: the NMR-CUFF, *Journal of magnetic resonance*, 208 (2011) 27-33.
- [22] E. Danieli, J. Perlo, B. Blümich, F. Casanova, Highly stable and finely tuned magnetic fields generated by permanent magnet assemblies, *Physical review letters*, 110 (2013) 180801.
- [23] K. Turek, P. Liszkowski, Magnetic field homogeneity perturbations in finite Halbach dipole magnets, *Journal of magnetic resonance*, 238 (2014) 52-62.

- [24] N. Ichijo, K. Takeda, K. Takegoshi, Paramagnetic shimming for wide-range variable-field NMR, *Journal of magnetic resonance*, 246 (2014) 57-61.
- [25] P. Blümler, F. Casanova, CHAPTER 5. Hardware Developments: Halbach Magnet Arrays, (2015) 133-157.
- [26] Q. Chen, G. Zhang, Y. Xu, X. Yang, Design and simulation of a multilayer Halbach magnet for NMR, *Concepts in Magnetic Resonance Part B: Magnetic Resonance Engineering*, 45 (2015) 134-141.
- [27] L.C. Barnsley, D. Carugo, J. Owen, E. Stride, Halbach arrays consisting of cubic elements optimised for high field gradients in magnetic drug targeting applications, *Physics in medicine and biology*, 60 (2015) 8303-8327.
- [28] Y. Xu, Q. Chen, G. Zhang, Z. Pei, X. Yang, Shim Coils Design for Halbach Magnet Based on Target Field Method, *Applied Magnetic Resonance*, 46 (2015) 823-836.
- [29] S. Utsuzawa, E. Fukushima, Unilateral NMR with a barrel magnet, *Journal of magnetic resonance*, 282 (2017) 104-113.
- [30] W. Lee, G. Cornilescu, H. Dashti, H.R. Eghbalnia, M. Tonelli, W.M. Westler, S.E. Butcher, K.A. Henzler-Wildman, J.L. Markley, Integrative NMR for biomolecular research, *Journal of biomolecular NMR*, 64 (2016) 307-332.
- [31] P. Blümler, F. Casanova, Hardware developments: Halbach magnet arrays, in M. Johns, E.O. Fridjonsson, S. Vogt, A. Haber (Eds.), *Mobile NMR and MRI: Developments and Applications*, Royal Chemical Society, Cambridge, 2015.
- [32] R.C. Jachmann, D.R. Trease, L.S. Bouchard, D. Sakellariou, R.W. Martin, R.D. Schlueter, T.F. Budinger, A. Pines, Multipole shimming of permanent magnets using harmonic corrector rings, *The Review of scientific instruments*, 78 (2007) 035115.
- [33] P. Nath, C.K. Chandrana, D. Dunkerley, J.A. Neal, D. Platts, The “Shim-a-ring” magnet: Configurable static magnetic fields using a ring magnet with a concentric ferromagnetic shim, *Applied Physics Letters*, 102 (2013) 202409.
- [34] F.X. Li, J.P. Voccio, M. Cheol Ahn, S. Hahn, J. Bascunan, Y. Iwasa, An Analytical Approach towards Passive Ferromagnetic Shimming Design for a High-Resolution NMR Magnet, *Superconductor science & technology*, 28 (2015).
- [35] A.J. Parker, W. Zia, C.W. Rehorn, B. Blümich, Shimming Halbach magnets utilizing genetic algorithms to profit from material imperfections, *Journal of magnetic resonance*, 265 (2016) 83-89.
- [36] W. Tang, W. Wang, W. Liu, Y. Ma, X. Tang, L. Xiao, J.H. Gao, A home-built digital optical MRI console using high-speed serial links, *Magnetic resonance in medicine*, 74 (2015) 578-588.
- [37] G. Giovannetti, V. Hartwig, V. Viti, G. Gaeta, R. Francesconi, L. Landini, A. Benassi, Application of undersampling technique for the design of an NMR signals digital receiver, *Concepts in Magnetic Resonance Part B: Magnetic Resonance Engineering*, 29B (2006) 107-114.
- [38] Z. Liu, C. Zhao, H. Zhou, H. Feng, A novel digital magnetic resonance imaging spectrometer, *Conference proceedings : ... Annual International Conference of the*

IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Annual Conference, 1 (2006) 280-283.

[39] W. He, X. Qin, R. Jiejing, L. Gengying, Four-channel magnetic resonance imaging receiver using frequency domain multiplexing, *The Review of scientific instruments*, 78 (2007) 015102.

[40] K. Takeda, OPENCORE NMR: open-source core modules for implementing an integrated FPGA-based NMR spectrometer, *Journal of magnetic resonance*, 192 (2008) 218-229.

[41] K. Takeda, A highly integrated FPGA-based nuclear magnetic resonance spectrometer, *The Review of scientific instruments*, 78 (2007) 033103.

[42] N. Ruipeng, D. Yidong, Y. Guang, L. Gengying, A digital receiver with fast frequency- and gain-switching capabilities for MRI systems, *Magma*, 22 (2009) 333-342.

[43] W. Tang, H. Sun, W. Wang, A digital receiver module with direct data acquisition for magnetic resonance imaging systems, *The Review of scientific instruments*, 83 (2012) 104701.

[44] A. Asfour, K. Raoof, J.P. Yonnet, Software Defined Radio (SDR) and Direct Digital Synthesizer (DDS) for NMR/MRI instruments at low-field, *Sensors*, 13 (2013) 16245-16262.

[45] D. Ha, J. Paulsen, N. Sun, Y.Q. Song, D. Ham, Scalable NMR spectroscopy with semiconductor chips, *Proceedings of the National Academy of Sciences of the United States of America*, 111 (2014) 11955-11960.

[46] L. Hong, D. Zu, Shimming Permanent Magnet Of MRI Scanner, *PIERS Online*, 3 (2007) 859-864.

[47] M.E. Lacey, R. Subramanian, D.L. Olson, A.G. Webb, J.V. Sweedler, High-Resolution NMR Spectroscopy of Sample Volumes from 1 nL to 10 μ L, *Chemical reviews*, 99 (1999) 3133-3152.

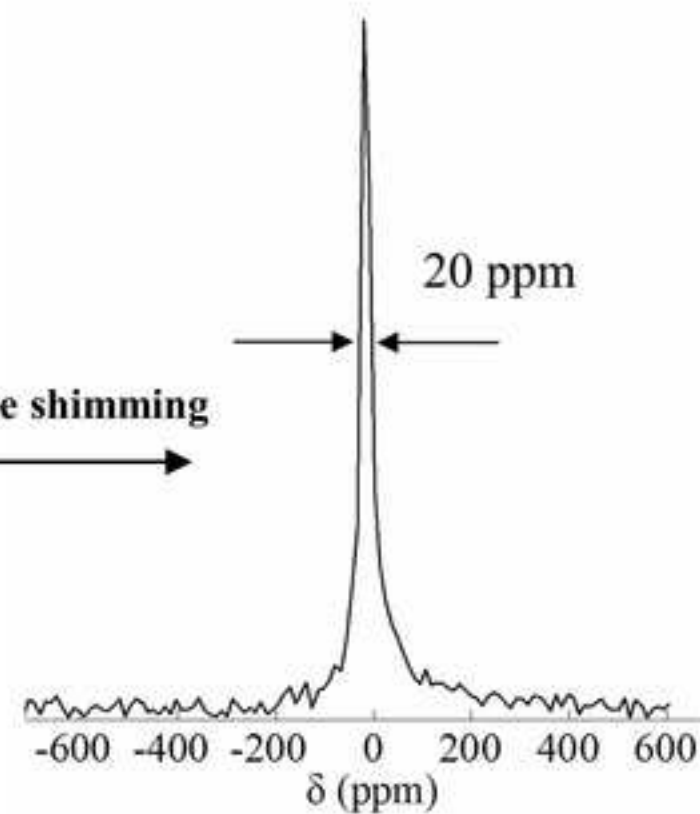


3-layer Hablach magnet



**Single board digital spectrometer
with better SNR**

Passive shimming



Highlights

- A low-cost desktop NMR system with 3-layer Halbach magnet was developed.
- A wedge-shaped mechanical structure was designed for magnet rapid assembling and a shimming manner with two magnetic blocks and steel pieces was proposed to reduce the cost and effort.
- Digital modulation and demodulation were utilized with a single FPGA and the spectrometer was realized on a single compact board for better SNR and compact size.